

ACP

Aarambh Chapterwise Problems



LINE DO NOT CROSS

Class-10th

How do organisms
reproduce?

How do Organisms Reproduce?

Read the question properly!

PART 1 (MCQs)

1. Which statement correctly distinguishes asexual reproduction from sexual reproduction?

- A. Asexual involves gametes ;sexual does not involve gametes .
- B. Both require fertilization
- C. Both produces identical offsprings .
- D. Asexual involves one parent ; sexual involves two parent .

2. Variations arise mainly due to

- A. DNA copying and sexual reproduction
- B. Only environmental change
- C. Growth and Nutrition
- D. Both A and B

3. Which is a key advantage of sexual reproduction ?

- A. Faster reproduction
- B. Cause variations which promote evolution
- C. No cell division occur
- D. Only one parent is needed

4. In which fission does the nucleus divide repeatedly before the cytoplasm ?

- A. Binary Fission
- B. Multiple Fission
- C. Budding
- D. Fragmentation

5. Identify the organism shown in the diagram and state its mode of reproduction.

- A. Hydra — Budding
- B. Planaria — Regeneration
- C. Spirogyra — Fragmentation
- D. Plasmodium — Multiple fission



6. Select from the following the conditions responsible for the rapid spread of bread mould on a slice of bread:

- (i) Formation of a large number of spores
- (ii) Presence of moisture and nutrients in bread
- (iii) Low temperature
- (iv) Presence of hyphae

Choose the correct option:

- (A) (i) and (ii)
- (B) (ii) and (iv)
- (C) (ii) and (iii)
- (D) (iii) and (iv)

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7. Select the option that correctly matches the organisms given in Column I with their modes of reproduction in Column II.

Column I	Column II
P. Leishmania	1. Regeneration
Q. Spirogyra	2. Multiple fission
R. Planaria	3. Binary fission
S. Plasmodium	4. Fragmentation

- A. P-4, Q-2, R-1, S-3
 B. P-1, Q-4, R-3, S-2
 C. P-3, Q-4, R-1, S-2
 D. P-4, Q-3, R-2, S-1

8. What is the function of the pollen tube?

- (a) Produce pollen
 (b) Carry the male gamete
 (c) Produce the ovule
 (d) Form the seed coat

9. The plants that can be raised by the method of vegetative propagation are:

- (a) Sugarcane, roses, grapes
 (b) Sugarcane, mustard, potato
 (c) Banana, orange, mustard
 (d) Papaya, mustard, potato

10. What happens to the following parts of a flower after fertilisation? Match the followings

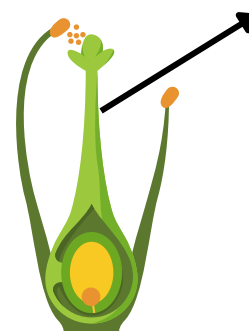
Column I	Column II
P. Ovary	1. Develops into an embryo
Q. Ovule	2. Develops into a fruit
R. Zygote	3. Develops into a seed
S. Petals and sepals	4. Usually shrivel and fall off

- A. P-2, Q-3, R-1, S-4
 B. P-3, Q-2, R-1, S-4
 C. P-2, Q-1, R-3, S-4
 D. P-1, Q-3, R-2, S-4

11. Identify part X.

It connects the stigma to the ovary and provides a pathway for pollen tube .

- A. Stigma
 B. Style
 C. Filament
 D. Ovary

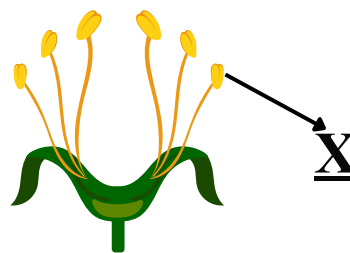


X

12. A student accidentally removes the labelled part X from a flower. As a result, pollen grains cannot be produced.

Identify X.

- A. Filament
- B. Anther
- C. Pollen tube
- D. Stigma



13. The table lists some changes that occur inside the female body after fertilization of an egg with sperm.

- A. Rhythmic contraction of uterus muscles for the baby's birth
- B. Formation of placenta.
- C. Implantation of embryo
- D. Development of the fetus.
- E. Cell division in zygote

Which option correctly sequences these events?

- (a) C→B→E→A→D
- (b) E→C→D→B→A
- (c) E→C→B→D→A
- (d) C→E→A→B→D

14. Which of the following statements regarding puberty is incorrect?

- A. In girls, eggs begin to mature after puberty.
- B. Several physical changes occur in the body during puberty.
- C. Sperm production begins at puberty in boys.
- D. Puberty occurs at exactly the same age in every individual.

15. A student says, “Menstruation and the menstrual cycle are the same because both are associated with monthly reproductive changes in females.” Which statement correctly explains the difference?

- A. The menstrual cycle refers only to the release of an ovum from the ovary.
- B. Menstruation occurs only when fertilisation takes place.
- C. Menstruation refers to the shedding of the thickened uterine lining when fertilisation does not occur, whereas the menstrual cycle includes the sequence of reproductive changes occurring during the cycle.
- D. Menstruation occurs because the ovaries permanently stop functioning.

16. Which of the following groups contains only sexually transmitted diseases (STDs)?

- (a) Tuberculosis , AIDS , Syphilis
- (b) AIDS ,Syphilis, Gonorrhoea
- (c) Malaria , Syphilis, Gonorrhoea
- (d) None of the above

17. Match the contraceptive methods in Column A with their correct description in column B

Column A	Column B
P. Oral pills	1. Prevent sperm from reaching the ovum
Q. Condoms	2. Vas deferens are cut or blocked
R. Copper-T	3. Placed in the uterus to prevent pregnancy
S. Vasectomy	4. Prevent ovulation

- A. P-4, Q-1, R-3, S-2
 B. P-1, Q-4, R-3, S-2
 C. P-3, Q-4, R-1, S-2
 D. P-4, Q-3, R-2, S-1

Prepare this section very carefully. most confusing section

Assertion - Reason

Directions : Choose the correct option-

- a) Both Assertion and Reason are true, and reason is the correct explanation of Assertion.
 b) Both Assertion and Reason are true, and reason is not the correct explanation of Assertion.
 c) Assertion is true, Reason is false.
 d) Assertion is false, Reason is true.

18. **Assertion (A):** Asexual reproduction generally produces offspring that are genetically very similar to the parent.

Reason (R): DNA copying occurs with high accuracy, although minor variations may arise during the copying process.

19. **Assertion (A):** DNA copying is necessary for the transmission of hereditary information during reproduction, but it does not ensure that the offspring will be an absolute identical copy of the parent.

Reason (R): The process of DNA copying can involve minor inaccuracies which may introduce variations.

20. **Assertion (A)** Regeneration and Reproduction are always the same process.

Reason (R): Regeneration involves regrowth of lost body parts , although in some organisms it can also lead to formation of new individuals

21. A pollen grain has germinated on the stigma and its pollen tube is growing towards the ovule.

Assertion: Fertilisation cannot yet be inferred merely from the observation described above.

Reason: Formation of a pollen tube precedes the fusion of the male and female gametes.

22 One fallopian tube of a woman is blocked, although the corresponding ovary continues to function normally.

Assertion: The blockage may prevent fertilisation involving the ovum released from that ovary.

Reason: Ovulation and fertilisation are separate events, and the fertilisation normally occurs in the Fallopian tube.

23 **Assertion:** Implantation cannot be considered the event that produces the zygote.

Reason: The zygote is formed by the fusion of male and female gametes, whereas implantation involves attachment of the developing embryo to the uterine lining.

24 . Four students are discussing reproduction.

Aarav: Asexual reproduction generally involves a single parent.

Meera: DNA copying is associated with the transmission of hereditary information.

Riya: Sexual reproduction can produce variation among offspring.

Kabir: Since DNA copying occurs before reproduction, every offspring must be genetically identical to the parents.

Who has made the least scientifically valid statement?

- (a) aarav
- (b) Meera
- (c) Riya
- (d) Kabir

Be straightforward with answers

Part 2 (SUBJECTIVE QUESTIONS)

Short answer type questions

25. After implantation a Specialised structure develops between the developing embryo and the uterine wall. It enables the developing embryo to obtain substances required for growth while also helping in the removal of certain waste products. Identify the structure and state any two functions. Why is the formation of this structure important after implantation?

26. Compare the reproductive parts of flowers and humans and answer the questions below:

- (a) Name the structure that produces male gametes in humans and the floral part that produces pollen grains in flowering plants.
- (b) Where does fertilisation occur in humans? Where does the fusion of male and female gametes occur in a flower?
- (c) State one similarity between sexual reproduction in humans and flowering plants.

27. What are the advantages of sexual reproduction over asexual reproduction? Explain the importance of variations produced during sexual reproduction.

28. How do Leishmania and Plasmodium reproduce, and mention the difference in their modes of reproduction?

29. Explain reproduction by spores in Rhizopus with the help of a labelled diagram. Why does Rhizopus not multiply on a dry slice of bread?

30. A farmer wants to grow disease-free banana plants quickly on a large scale even though the variety produces very few viable seeds. Which method from the chapter should be used and why? Explain the process briefly.

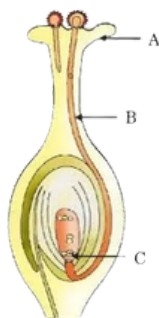
31. What happens when:

- (1) Leaves of Bryophyllum fall on the soil?
- (2) Planaria is cut into many pieces.
- (3) Sporangia of Rhizopus on maturation liberate spores? Mention the modes of reproduction in each of the above three cases.

32. What are sexually transmitted diseases (STDs)? Name one STD caused by a virus and one caused by a bacterium. Name any two contraceptive methods and state one advantage of each.

33.(i) Differentiate between self-pollination and cross-pollination. Write two advantages of cross-pollination.

(ii) Identify A, B, and C in the diagram given below and write one function of each.



34. A pollen grain lands on the stigma of a flower and subsequently develops a pollen tube. A student concluded that since the pollen has reached the stigma and the pollen tube has started growing, fertilization has already occurred. Do you agree? Give a reason for your answer.

35. Flowers and humans have structurally different reproductive systems, but some reproductive functions can be compared.

- Which part of a flower receives pollen? What reproductive function does it perform?
- Sperm is deposited into the female reproductive tract in humans. Why would it be incorrect to identify a particular human structure as the anatomical equivalent of the stigma merely because both are involved in receiving reproductive material?
- What does this comparison demonstrate about the compared reproductive structures of different organisms?

36. A human female does not experience fertilization during a particular reproductive cycle. What happens to the thickened lining of the uterus? Name the process associated with its shedding.

37. What is puberty? List any two changes that occur in boys and any two changes that occur in girls during puberty.

38. Why can regeneration not be considered the same as reproduction in all organisms? Explain with a suitable example.

39. Consider the following situations:

P: A couple wants a temporary contraceptive method that acts as a physical barrier to sperm.

Q: A woman undergoes a permanent contraceptive procedure in which the Fallopian tubes are cut or blocked

R: A person wants to use a method that can also help reduce the risk of transmission of certain sexually transmitted infections.

- Identify the appropriate method of P.
- Identify the procedure represented by Q and state how it prevents pregnancy.
- Identify the method appropriate for R and state why it is relevant to STI prevention.

40. A gardener wants to produce several new rose plants having the same desirable characteristics as the parent plant.

- Which mode of reproduction would be suitable?
- Why would the new plants generally resemble the parent plant?
- State one advantage of using vegetative propagation in such plants

41. A sperm enters the female reproductive tract but does not immediately encounter the ovum. It has to travel through the reproductive tract before reaching the region where fertilization normally takes place.

- 1) Name the usual site of fertilization in humans.
- 2) State the sequence of the major structures through which sperm passes before reaching this site, beginning from the point of entry.

42. A student writes the following sequence in her notebook:

Pollen deposition → Fertilisation → Zygote → Fruit

Her teacher marks the sequence incomplete.

- (a) What important event occurs after a pollen grain lands on the stigma and before fertilisation?
- (b) Where does the male gamete fuse with the female gamete in the flower?
- (c) What is formed immediately after the fusion of the male and female gametes?
- (d) Which floral structure develops into the fruit after fertilisation?

*Read the ques properly before
answering*

Long answer type questions

43. A farmer wants thousands of plants of a high-yielding banana variety that produces little viable seed.

- a) Which mode of reproduction would be more suitable?
- b) Name the technique used for rapid multiplication from small pieces of plant tissue.
- c) Why will the new plants generally retain the desired characteristics of the parent?
- d) Why is this method preferable to seed production in this case?
- e) If all plants are genetically very similar, why could a single disease become a serious problem for the crop?

44. A student says, Fertilization occurs in the uterus because the embryo develops there.

- a) Where does fertilization normally occur in humans?
- b) What is formed immediately after fertilization?
- c) Why is implantation different from fertilization?
- d) What role does the placenta play after implantation?

45. A species has 48 chromosomes in each of its body cells. A student claims that its gametes should also contain 48 chromosomes.

- (a) How many chromosomes should each gamete normally contain?
- (b) Why must gametes contain half the chromosome number of body cells?
- (c) Suppose the chromosome number were not reduced during gamete formation. Predict what would happen to the chromosome number after fertilisation and over successive generations.

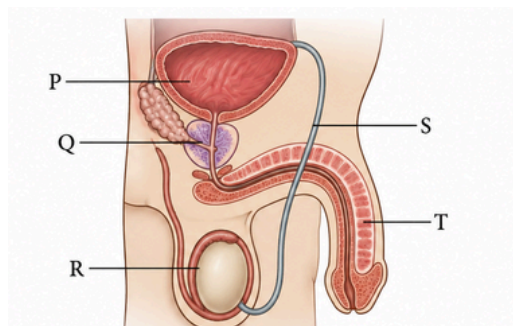
46. What are testes? List two functions performed by testes in human beings. Why are they located outside the abdominal cavity?

47. Write the functions of the following in the human female reproductive system: (a) Ovary (b) Oviduct (c) Uterus

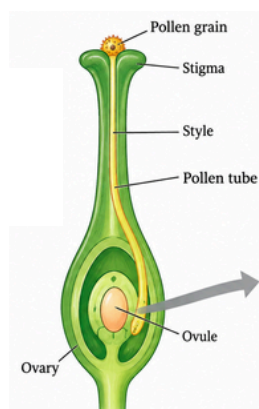
48. What is placenta? State its functions in human female. How does placenta help in the nourishment of the embryo?

49. A diagram of the human male reproductive system is provided with labels P, Q, R, S and T.

- Identify the structure labelled R which is responsible for production of male gametes.
- Starting from the testes, write the correct sequence of major structure through which sperm passes before leaving the body.
- A student states, The testes produce sperm as well as all the fluid that forms semen. Is it correct? Justify?
- Name one gland that contributes fluid to semen and state one function of the fluid secreted by it.

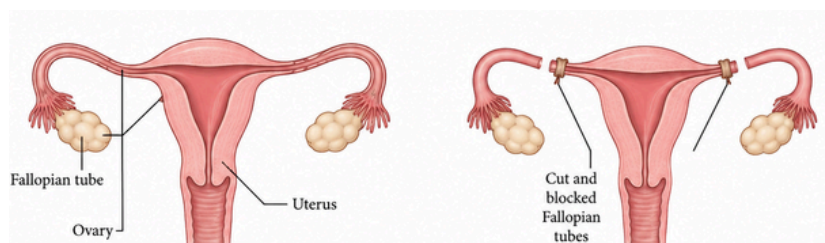


50. The diagram shows a pollen grain germinating on the stigma and a pollen tube growing towards an ovule. Answer the following questions.



- What is the function of the pollen tube?
- Which cells are carried through the pollen tube?
- Where does the male gamete fuse with the female gamete?
- What is formed immediately after this fusion?
- What happens to the ovule and ovary after fertilisation?

51. A diagram shows a contraceptive procedure in which the Fallopian tubes are cut/blocked.



- Identify the contraceptive procedure shown.
- Which two reproductive cells are prevented from meeting as a result of this procedure?
- Explain how blocking the Fallopian tubes prevents pregnancy without stopping the ovaries from functioning.
- Does this procedure provide protection against sexually transmitted infections (STIs) like condoms do? Give a reason.
- Is this method generally considered temporary or permanent?

PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

52. Puneet wants to grow banana plants.

- (i) Based on your knowledge of plant reproduction, should he use seeds or an alternative method of reproduction? Justify your answer.
- (ii) Offspring produced from a banana plant by vegetative propagation usually show very little variation. What causes variations, and can variations be beneficial, harmful or neutral? Explain.

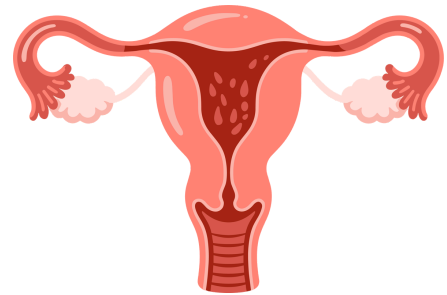
53. Annie was conducting research on the number of fruits produced by watermelon under different conditions. She grew 25 watermelon plants each in both glass house A and B. She introduced pollinators in glass house A only.

- (i) What difference will she observe in the number of fruits produced in the two glass houses? Explain with reason.
- (ii) List 3 changes that will occur in a flower once it gets fertilized.

54. During each menstrual cycle, one egg may mature and the uterus prepares a thick lining to receive a fertilised egg. If fertilisation occurs, the embryo gets implanted in the uterine lining and pregnancy begins. If fertilisation does not occur, the thickened lining breaks down and is removed from the body as menstrual flow.

i) The release of an egg from the ovary is called:

- (a) Fertilisation
- (b) Ovulation
- (c) Implantation
- (d) Menstruation



ii) Implantation normally occurs in the:

- (a) Ovary
- (b) Oviduct
- (c) Uterus
- (d) Vagina

iii) If fertilisation does not occur, the uterine lining:

- (a) Becomes thicker permanently
- (b) Breaks down
- (c) Produces sperm
- (d) Forms an embryo

iv) Why does the uterus prepare its lining every month?

- Basic Reproduction & Sexual vs Asexual — Q1, Q2, Q3, Q18, Q19, Q20, Q24, Q27, Q38, Q52(ii)
- DNA Copying & Variation — Q2, Q18, Q19, Q24, Q27, Q52(ii)
- Fission — Q4, Q7, Q28
- Leishmania & Plasmodium — Q7, Q28
- Budding — Q5
- Regeneration — Q7, Q20, Q31(2), Q38
- Fragmentation — Q7
- Spore Formation — Q6, Q29, Q31(3)
- Vegetative Propagation — Q9, Q30, Q31(1), Q40, Q43, Q52(i)
- Tissue Culture — Q30, Q43
- Flower Parts & Reproduction — Q10, Q11, Q12, Q26, Q33(ii), Q35(a), Q42, Q50
- Pollination — Q33(i), Q34, Q35, Q53(i)
- Pollen Tube & Fertilisation — Q8, Q21, Q26(b), Q33(ii), Q34, Q42, Q50
- Seed & Fruit Formation — Q10, Q42, Q50, Q53(ii)
- Human Reproductive System — Q13, Q22, Q23, Q25, Q26, Q41, Q44, Q46–Q49
- Puberty & Menstruation — Q14, Q15, Q36, Q37, Q54
- Contraception & STDs — Q16, Q17, Q32, Q39, Q51
- Chromosome Number & Gametes — Q45

SOLUTIONS



MULTIPLE CHOICE QUESTIONS [MCQ's]

- | | | |
|-------------|--------------|--------------|
| 1. D | 10. A | 18. A |
| 2. A | 11. B | 19. A |
| 3. B | 12. B | 20. D |
| 4. B | 13. C | 21. A |
| 5. A | 14. D | 22. A |
| 6. A | 15. C | 23. A |
| 7. C | 16. B | 24. D |
| 8. B | 17. A | |
| 9. A | | |

SHORT ANSWER TYPE QUESTIONS:

25. Placenta - The specialised structure is called the placenta.

It supplies the developing embryo with oxygen and nutrients and removes carbon dioxide and other wastes.

It is important because it provides a connection between the mother and embryo for the exchange of necessary substances.

26. Reproduction in flowers and humans

(a) Humans: Testes produce male gametes called sperms.

Flowering plants: Anthers produce pollen grains containing the male gametes.

(b) In humans, fertilisation normally occurs in the Fallopian tube or oviduct.

In flowering plants, the male gamete fuses with the female gamete inside the ovule.

(c) In both humans and flowering plants, sexual reproduction involves the fusion of male and female gametes to form a zygote.

27. Sexual reproduction produces greater variation among offspring because genetic material from two parents is combined.

These variations increase diversity within a species and may help some individuals survive when environmental conditions change.

Variations also provide the basis for evolution.

28. Leishmania and Plasmodium

Leishmania reproduces by binary fission, in which one cell divides into two daughter cells.

Plasmodium reproduces by multiple fission, in which one cell divides to form many daughter cells.

Thus, the main difference is the number of daughter cells produced.

29. Reproduction by spores in Rhizopus

Rhizopus forms spores inside a sporangium. When the sporangium bursts, spores spread through the air and grow into new Rhizopus under suitable conditions.

Labels for diagram: Sporangium, spores, sporangiophore, and hyphae.

Rhizopus does not multiply well on dry bread because its spores require moisture to germinate and grow.

30. The farmer should use tissue culture (vegetative propagation).

A small piece of healthy banana plant tissue is grown in a nutrient medium under sterile conditions.

Many disease-free plants can be produced quickly from one healthy plant.

It is useful when the plant does not produce seeds.

31. What happens when:

(1) Bryophyllum leaves fall on soil: New plants develop from the buds on the leaf margins. → Vegetative propagation.

(2) Planaria is cut into pieces: Each piece can grow into a complete organism. → Regeneration.

(3) Rhizopus sporangia burst: Spores are released and spread through air. → Spore formation.

32. Sexually transmitted diseases (STDs) are diseases that spread mainly through sexual contact.

Viral STD: HIV/AIDS

Bacterial STD: Gonorrhoea

Condom: Acts as a physical barrier and also helps reduce the risk of transmission of many STDs.

Oral contraceptive pills: Prevent ovulation and provide a temporary, non-surgical method of contraception.

33. (i) Advantages of cross-pollination:

- It produces greater genetic variation among offspring.
- These variations may help a population survive changing environmental conditions.

(ii) Diagram:

- A – Stigma: receives pollen grains
- B – Pollen tube: carries male gametes towards the ovule.

- C – Ovule: contains the female gamete and develops into a seed after fertilisation.

34. Pollen tube and fertilisation

No, I do not agree.

The growth of a pollen tube only shows that the male gametes are travelling towards the ovule.

Fertilisation occurs when a male gamete fuses with the female gamete inside the ovule.

35. Comparison of reproductive structures

- (a) The stigma receives pollen grains and helps in the beginning of pollen-tube growth.
- (b) It is incorrect to call a particular human structure the equivalent of the stigma just because both receive reproductive material; their structures and functions are different.
- (c) This shows that reproductive functions may be similar, but the structures performing them can be different in different organisms.

36. Uterine lining

If fertilisation does not occur, the thickened lining of the uterus is no longer needed.

It breaks down and is removed from the body along with blood and tissue.

This process is called menstruation.

37. Puberty is the stage during which the body undergoes changes that lead to sexual maturity and the reproductive organs become functional.

Changes in boys:

- Growth of facial and body hair.
- Voice becomes deeper.

Changes in girls:

- Breasts develop.
- Eggs begin to mature and menstruation starts.

38. Regeneration is the ability of an organism to regrow lost or damaged body parts.

In some simple organisms such as Planaria, if the body is cut into several pieces, each piece can regenerate into a complete organism.

However, regeneration cannot be considered the same as reproduction in all organisms because most organisms do not normally reproduce by being cut into pieces.

39. Contraceptive methods

- (a) P: Condom — it acts as a physical barrier and prevents sperm from reaching the egg.
- (b) Q: Tubectomy — the fallopian tubes are cut and blocked, preventing the sperm and egg from meeting.
- (c) R: Condom — it helps prevent pregnancy and also reduces the risk of many sexually transmitted infections (STIs)

40. (a) Vegetative propagation, such as stem cutting, would be suitable.
- (b) The new plants are produced from a single parent without the fusion of male and female gametes. Therefore, they are genetically very similar to the parent and retain its desirable characteristics.
- (c) One advantage is that many plants with the desired characteristics of the parent can be produced quickly.

41. Fertilisation in humans

1. The usual site of fertilisation is the fallopian tube (oviduct).
 2. Sperm passes through: Vagina → Cervix → Uterus → Fallopian tube.
- It then reaches the fallopian tube, where it may meet the ovum and fertilisation can occur.

- 42.(a) The pollen grain germinates on the stigma and forms a pollen tube, which grows through the style towards the ovule.
- (b) The male gamete fuses with the female gamete inside the ovule.
- (c) A zygote is formed immediately after the fusion of the gametes.
- (d) The ovary develops into the fruit after fertilisation.

LONG ANSWER TYPE QUESTIONS:

43. Banana plants

- (a) Asexual reproduction (vegetative propagation) is more suitable.
- Banana produces very few viable seeds, so growing from seeds is difficult.
 - Asexual reproduction can produce many plants quickly.
 - The new plants will also have the same useful features as the parent plant.
- (b) Technique used for rapid multiplication
- The technique is called tissue culture or micropropagation.
 - A small piece of healthy plant tissue is taken from the parent plant.
 - It is grown in a nutrient-rich medium under sterile conditions.
 - The tissue develops into small plantlets, which are later transferred to soil.
- (c) Tissue culture is a form of asexual reproduction.
- Only one parent is involved in producing the new plants.
 - There is no fusion of male and female gametes.
 - Therefore, the new plants are genetically very similar to the parent and retain its desired characteristics.
- (d) The banana variety produces very few viable seeds, so seed production is not reliable.
- Tissue culture can produce thousands of plants in a short time.
 - The plants are generally uniform and retain the desired high-yielding characters.
 - Healthy, disease-free plants can also be produced under sterile conditions.
- e) The plants produced by tissue culture are genetically very similar to one another.
- Therefore, they have almost the same resistance or weakness to a disease.
 - If a disease can infect one plant, it may infect many or all plants.
 - This lack of genetic variation can cause large crop losses.

44. Fertilisation and implantation

(a) Fertilisation normally occurs in the fallopian tube (oviduct).

- A sperm meets and fuses with the ovum in the fallopian tube.
- This fusion is called fertilisation.
- It results in the formation of a zygote.

(b) A zygote is formed immediately after fertilisation.

- It is formed by the fusion of the male gamete (sperm) and female gamete (ovum).
- The zygote contains genetic material from both parents.
- It then begins to divide and develop into an embryo.

(c) Fertilisation is the fusion of sperm and ovum to form a zygote.

- It normally occurs in the fallopian tube.
- Implantation is the attachment of the developing embryo to the lining of the uterus.
- Thus, fertilisation and implantation are two different events occurring at different places.

(d) Role of placenta after implantation

- The placenta develops between the embryo and the uterine wall.
- It supplies the developing embryo with oxygen and nutrients from the mother.
- It also carries carbon dioxide and other waste products away from the embryo.
- Thus, the placenta helps the embryo grow and develop inside the uterus.

45. Chromosome number

(a) How many chromosomes should each gamete contain?

The body cells contain 48 chromosomes.

Therefore, each gamete should contain 24 chromosomes.

Gametes are haploid, meaning they contain half the chromosome number.

So, the correct answer is 24 chromosomes.

(b) During fertilisation, one male gamete fuses with one female gamete.

If both contained 48 chromosomes, the zygote would have 96 chromosomes.

Instead, each gamete has 24 chromosomes.

Their fusion restores the normal body-cell number of 48 chromosomes.

(c) If chromosome number were not reduced during gamete formation, each gamete would contain 48 chromosomes. Fusion of two such gametes would form a zygote with 96 chromosomes.

If chromosome reduction again failed in the next generation, gametes would contain 96 chromosomes and their fusion would produce a zygote with 192 chromosomes. Thus, the chromosome number would keep doubling in successive generations.

46. Testes are a pair of male reproductive organs.

- They perform two main functions:
- Produce sperms.
- Produce the male sex hormone testosterone.
- The testes are located outside the abdominal cavity in the scrotum because sperm formation requires a temperature lower than normal body temperature.

47. Functions of female reproductive organs

- (a) Ovary: It produces ova (eggs) and female hormones such as oestrogen and progesterone.
- (b) Oviduct/Fallopian tube: It carries the ovum towards the uterus and is usually the site of fertilisation. It also provides the pathway through which the ovum travels after ovulation
- (c) Uterus: It is the organ where the embryo implants and develops.
- These organs work together to support reproduction and development of the baby.

48. Placenta

The placenta is a specialised structure formed between the developing embryo and the wall of the uterus. It supplies the embryo with oxygen and nutrients from the mother's blood. It also carries carbon dioxide and other waste products away from the embryo. The placenta is connected to the embryo by the umbilical cord. Thus, it plays an important role in the growth and nourishment of the embryo

49. Human male reproductive system

- (a) R represents the testis.

The testes are the male reproductive organs located in the scrotum. They produce sperms by the process of sperm formation. They also secrete the hormone testosterone. Testosterone is responsible for the development of male secondary sexual characters.

- (b) Path of sperm

Starting from the testes, sperm passes through the vas deferens (sperm duct). It then joins the ducts of the reproductive glands and enters the urethra. The urethra carries semen towards the outside of the body. Therefore, the main pathway is: Testes → Vas deferens → Urethra → Penis → Outside. The sperm mixes with fluids from glands to form semen.

- (c) No, this statement is incorrect.

Testes produce sperms, but they do not produce all the fluid present in semen. The fluid is mainly added by glands such as the seminal vesicles and prostate gland. Semen therefore contains sperms plus glandular fluids. These fluids provide nourishment and help sperm movement.

- (d) The seminal vesicles are glands that add fluid to the semen.

Their secretion contains nutrients such as fructose. This provides energy to the sperms for movement. The prostate gland also adds fluid that helps sperm function. Thus, reproductive glands help form semen and support sperm survival.

50. Pollen tube and fertilisation

- (a) The pollen tube carries the male gametes from the pollen grain towards the ovule.
- (b) Male gametes or male germ cells are carried through the pollen tube.
- (c) The male gamete fuses with the female gamete inside the ovule.
- (d) A zygote is formed immediately after fusion.
- (e) After fertilisation:
- Ovule → Seed
 - Ovary → Fruit

51. (a) Tubectomy.

(b) Sperm and ovum are prevented from meeting.

(c) The Fallopian tubes are cut or blocked, so sperm cannot reach the ovum. The ovaries continue to release ova and produce hormones normally.

(d) No. Tubectomy does not protect against STIs because it does not prevent contact with infected body fluids. Condoms can reduce the risk of many STIs.

(e) It is generally considered a permanent method of contraception.

CASE STUDY/SOURCE BASED QUESTION:

52.(i)Puneet should use an alternate method such as vegetative propagation instead of seeds.

Reason: Banana is commonly propagated vegetatively. It produces plants quickly, and the new plants are genetically similar to the parent, retaining desirable characteristics.

(ii)Variation arises mainly due to DNA copying errors and sexual reproduction.

Variations can be good, bad, or neutral depending on the environment. They are important because useful variations can help organisms survive changing environmental conditions.

53.(i)Glass house A will produce more fruits than glass house B.

Reason: Pollinators help transfer pollen from the anther to the stigma. This increases pollination and hence the chances of fertilisation and fruit formation. In B, the absence of pollinators will reduce pollination and fruit production.

(ii)Three changes after fertilisation are:

- 1.Ovary develops into a fruit.
- 2.Ovules develop into seeds.
- 3.Petals, sepals and other floral parts usually wither and fall off.

54. (i) B — Ovulation

(ii) C — Uterus

(iii) B — Breaks down

(iv) The uterus prepares a thick lining every month to provide a suitable site for implantation and nourishment of the embryo if fertilisation occurs.